

Transfer Learning And Cross-Linguistic Generalization In Multilingual Hate Speech Detection: Approaches And Challenges

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Abstract— Detecting hate speech in multilingual environments remains a critical challenge, particularly in low-resource and unseen languages during training. This study presents a systematic review of approximately 44 research papers, analyzing transfer learning strategies, cross-lingual generalization, and the challenges associated with linguistic biases and data scarcity. The study identifies key research gaps, including limitations in knowledge transfer between languages, biases in large language models (LLM), and inconsistencies in the classification of hate speech between linguistic contexts. The primary contribution of this work lies in its comprehensive analysis of existing methodologies, identification of critical shortcomings, the effectiveness of multilingual hate speech detection models. This study advocates for the development of more equitable and efficient models by improving cross-lingual knowledge transfer, mitigating biases, and strengthening detection capabilities in low-resource languages, ultimately contributing to safer and more inclusive digital environments.

Keywords— NLP, Hate Detection, Multilingual Model, Cross-Language, Offensive Speech, zero-shot-learning, Transfer learning

I. INTRODUCTION

Hate speech (HS), defined as discriminatory or violent language targeting individuals/groups based on race, religion, nationality, or identity [3], has become a growing global concern. The lack of a universal definition makes it challenging to distinguish HS from related concepts like cyberbullying, toxicity, and offensive language [1,4,5]. Effective HS detection is crucial for maintaining safe online spaces, protecting vulnerable communities, and ensuring legal compliance [6,7]. Early detection systems (2000s) primarily focused on English, using limited keyword-based approaches [6,8]. Current challenges include definition inconsistencies across datasets and resource disparities between languages (e.g., 25 English vs. 4 German datasets) [8].

Natural language processing (NLP) has become a powerful tool in detecting harmful content (HS) in texts. Techniques like sentiment analysis, information mining, and machine learning have improved the effectiveness of HS detection in multilingual environments. However, these models were often focused on a single language, making them ineffective in multilingual environments. As societies became more linguistically diverse, research focused on adapting NLP models to different languages. Techniques like deep learning and neural networks have improved performance in detecting HS across languages, allowing for expanded applications in social media and news. Multilingual HS detection is a growing field that addresses the complexities of identifying harmful content across diverse languages and cultural contexts. The

knowledge gained from this can be used to develop targeted interventions to more effectively address HS [4].

HS detection can be improved by using multilingual methods, utilizing transfer learning principles. This involves transferring knowledge from a high-resource language like English to a low-resource language. Multilingual models can be built using corpora from multiple languages and applied to detect HS in a target language. Research shows that multilingual models can outperform monolingual models and can detect HS in a target language without a corpus from the target language [2]. Large Language Models (LLMs) have significantly improved hate speech detection, but challenges remain, such as addressing biases in training data and models. Ethical considerations and continuous efforts to mitigate bias are crucial [9].

Natural Language Processing (NLP) employs multi-lingual and cross-lingual approaches to understand languages different from English. Multi-lingual NLP involves developing models for simultaneous processing of multiple languages, while cross-lingual NLP focuses on interlanguage communication and comprehension, transferring information or models from abundant languages to limited ones, using machine translation, word alignment, parallel corpora, and bilingual dictionaries. Although the term "cross-lingual learning" usually refers to the use of CLTL techniques, the terms "multilingual learning" and "cross-lingual learning" are occasionally used interchangeably [10]. Code-mixed content, or a combination of languages in a single sentence, can be found in some multilingual data [11]. We take a more comprehensive view of cross-lingual learning, even though some researchers make comparisons between these concepts and the connection between transfer learning and multi-task learning [12]. The process of transferring knowledge between languages is what we refer to as cross-lingual learning, which is a concept that naturally encompasses multilingual learning.

Multilingual HS detection is a promising solution for non-English language HS detection. This survey explores the detection of HS in multilingual environments using different techniques. We survey research from 2018 to 2024, distributing studies based on multilingual, cross-lingual the focus is on transfer learning for low-resource languages that were not known during training and mixed-language HS detection, examining early research, identifying gaps, and highlighting improvements in existing models. The study analyzes multilingual datasets, assesses current challenges and potential solutions, and aims to identify future research directions in multilingual HS.

Structure of the Paper: The paper begins with a research methodology section detailing the selection criteria for the surveyed papers. This is followed by an overview of the methodologies employed and the corresponding studies under each approach. Subsequent sections explore multilingual hate speech detection and cross-linguistic studies, as well as a summary of key hate speech datasets in various languages. The

discussion then shifts to challenges and opportunities in the field, leading to future research directions. Finally, the paper concludes with a summary of key insights.

II. METHODOLOGY

This study systematically evaluates multilingual and cross-lingual hate speech detection (HSD) methodologies, focusing on knowledge transfer from high- to low-resource languages. Following strict inclusion criteria (peer-reviewed studies from 2018-2024 addressing cross-lingual/zero-shot HSD with empirical metrics) and exclusion criteria (monolingual studies, theoretical papers, duplicates), we searched IEEE Xplore, ACM Digital Library, Springer, and Google Scholar using keywords like "cross-lingual hate speech detection" and "multilingual transfer learning." Selected studies were categorized into multilingual HSD (simultaneous language processing) and cross-lingual HSD (unseen language generalization), then analyzed for detection methodologies (model architectures, transfer approaches), dataset characteristics (languages, sizes, annotation schemes), performance outcomes (F1-score, accuracy), and identified research gaps. This structured review provides a comprehensive assessment of current techniques while highlighting future directions in multilingual HSD research. Figure 1 shows the distribution of the percentages of use of different languages in multilingual hate speech detection research or systems, and Figure 2 shows the distribution of the number of research papers on detecting multilingual and cross-lingual hate speech during the years 2018 to 2025.

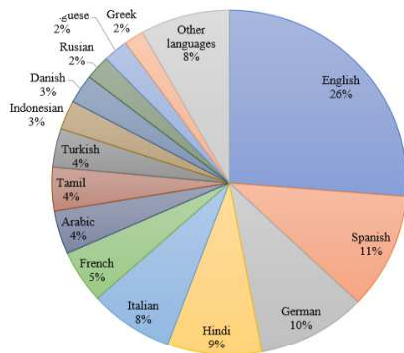


Fig. 1. : Researchers' focus on languages in multilingual hate speech

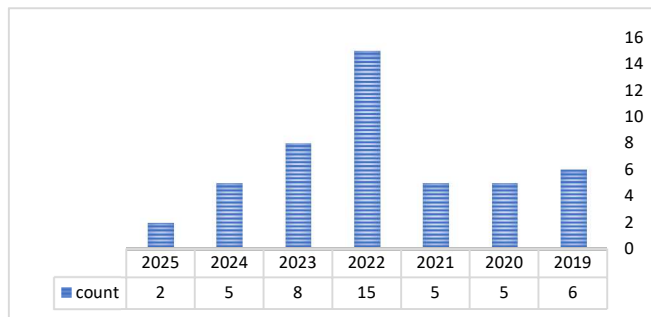


Fig. 2. Interest in detecting multilingual and cross-linguistic hate speech over the years

III. APPROACHES ON MULTILINGUAL HATE SPEECH DETECTION

Research on multilingual HS detection has shifted from monolingual models to cross-lingual and multilingual strategies. Techniques like zero-shot learning and few-shot learning are used to detect offensive language in low-resource languages. Current approaches include traditional machine learning, deep neural networks, transfer learning, machine

translation, ensemble learning, meta-learning, multi-task learning, and unsupervised learning. This study provides an overview of existing approaches for multilingual and cross-lingual HS detection.

A. Type of Model

1) Machine learning methods

Machine learning methods like logistic regression effectively detect offensive language in multiple languages, with deep neural networks and transfer learning methods being more effective. Traditional techniques like naive Bayes, k-nearest neighbors, decision trees, random forests, and support vector machines can still be considered.

A multilingual system for detecting hate speech and offensive language was developed by [13, 17–20] and demonstrated online retraining capability. Cross-linguistic experiments by [21] found that the laser model and the LR model were the most effective combined models.

2) Deep learning methods

Deep neural networks (DNNs) have been widely used for multilingual offensive language recognition due to their ability to learn complex textual representations across different languages. Convolutional neural networks (CNNs), CNN-GRU, and recurrent neural networks (RNNs) such as long short-term memory (LSTM) have been widely used. LSTM has been used for word embedding, while bidirectional LSTM (BiLSTM) has been implemented by several researchers. Researchers in [22] proposed a hate speech detection model called CCNL-Ex, which uses a multilingual capsule network learning (CCNL) approach with two parallel architectures for source and target languages. Other researches using deep learning are [8, 17, 19, 23–31].

3) Transfer Learning (TL)

Multilingual Language Models (MLMs) are powerful tools for Natural Language Processing (NLP) tasks, using shared representations and training techniques. Models like BERT, mBERT, and XLM-RoBERTa, built on Transformer architecture, capture linguistic nuances and improve performance for low-resource languages, while RoBERTa, DeBERTa, and DistilBERT offer optimized global applications.

Through our survey, researchers who have used multilingual transformers in cross-language transfer have found that the best model is XLM-R [34]. Cross-lingual transfer learning is a promising approach for HS detection, particularly in low-resource languages where labeled datasets are scarce. By leveraging existing data from high-resource languages, researchers have developed methods that utilize bilingual word embeddings and deep-learning models to classify HS effectively. For instance, studies have shown that training in a source language and applying the model to a target language can yield good performance, even without labeled examples in the target language [35]. Techniques such as data augmentation and cross-lingual contrastive learning further enhance detection capabilities [36]. Additionally, addressing challenges like label imbalance through under sampling and oversampling techniques has proven beneficial [8]. The development of multilingual domain-specific word embeddings also contributes to improved classification results by capturing unique patterns of HS across different languages [8]. Overall, these advancements highlight the effectiveness of cross-lingual transfer learning in combating online HS. In our survey of businesses that used multilingual converters [14–16, 18, 19, 27, 28, 31, 33, 37–55].

B. Transfer Strategy

1) Multi-task learning (MTL)

Multi-task learning (MTL) is an increasingly popular deep learning approach that enhances model generalization by simultaneously training on multiple tasks, leveraging shared layers and parameters across NLP [56] [57]. Multi-task learning has emerged as a powerful approach for

detecting HS and aggression on social media platforms. Recent studies propose sophisticated frameworks that leverage shared representations to enhance detection accuracy. For instance, a transformer-based multi-task framework effectively identifies various forms of aggression and HS by simultaneously addressing multiple tasks, such as aggression identification and the classification of hate-offensive content [29, 58]. Additionally, the integration of emotional features into detection models has been shown to improve performance, as emotional knowledge aids in recognizing abusive content more reliably [59]. Furthermore, advanced models utilizing multi-layered approaches and ensemble learning have demonstrated superior results in detecting HS across different modalities, including text and images [60]. Multi-task learning enhances model generalizability and resistance to overfitting by training and sharing representations on multiple auxiliary tasks simultaneously across different languages or similar NLP tasks [61].

2) *Meta-learning (Meta-L):*

Meta-learning can be used in various methodologies within the machine learning realm, such as in automated machine learning (AutoML), few-shot learning, recommendation engines, and transfer learning [62]. Meta-learning has emerged as a powerful approach to the detection of HS, particularly in addressing the challenges posed by low-resource languages and limited datasets. Research has shown that meta-learning frameworks, such as Hate MAML, effectively utilize self-supervision strategies to enhance model performance in multilingual settings, enabling rapid adaptation to unseen languages and datasets [49]. Additionally, the systematic review of cross-lingual transfer learning techniques emphasizes the importance of diverse multilingual datasets and the need for innovative transfer strategies, such as instance, feature, and parameter transfer, to tackle the complexities of offensive language detection in social media[48–50].

3) *Ensemble Learning:*

Ensemble learning enhances model performance by combining multiple models, achieving greater accuracy than individual ones. It employs techniques like bagging (training on diverse data samples), boosting (improving weak models), and stacking (meta-modeling predictions), reducing errors and improving generalization [8,53] demonstrate its effectiveness.

4) *Unsupervised Learning (UL):*

Unsupervised learning offers promising methods for HS detection, particularly in environments where labeled data is limited. Unsupervised embeddings using Graph Auto-Encoders (GAE) present a promising approach for multi-domain and multilingual HS detection, [64]. By representing texts as nodes in a graph and employing transformer and convolutional layers, GAE generates low-dimensional embeddings that facilitate the classification of HS without labeled data [64]. This method complements existing multilingual HS detection efforts, which utilize language-agnostic embeddings like LASER and MUSE to create a generalized definition of HS across diverse datasets [16]. Furthermore,[54] presented an unsupervised model that uses transfer learning, sample generation, and cross-lingual mapping.

5) *Machine Translation and Transliteration Tools*

Translation and transliteration enable cross-lingual content conversion but face challenges in preserving cultural nuances, especially for offensive language detection. Tools like Google Translate, mT5, and AI4Bharat Transliteration support these processes, though accuracy varies for slurs and culturally specific terms. Models like TIF-DNN and AraBERT/MarBERT-based systems leverage these techniques, combining translation, transliteration, and multi-task learning to improve multilingual hate speech detection, outperforming baseline methods [56].

C. *Transfer Scenario*

1) *Zero-Shot Transfer*

Zero-Shot Transfer is a promising strategy for this purpose, where the model is trained on labeled source data from one or more languages, excluding the target language, and tested on the unseen target language. There are 25 papers under this transfer model, including subtypes such as single-source-to-single-target [34, 37, 38, 42–44, 47, 51, 61, 65, 66], multiple-source-to-single-target [13, 39, 61], pseudo-target boosted training [13, 22, 45], and parameter frozen [43].

2) *Joint learning*

involves training a model between languages using multiple channels, tasks, or objectives simultaneously. Parameter sharing occurs when shared parameters are shared across languages, allowing knowledge transfer to occur on shared parameters. This process can be categorized into three primary fusion stages: early fusion, model-level fusion, and late fusion. Early fusion involves a combination of datasets or embeddings from both source and target languages, with the entire model sharing parameters. This is commonly used in few-shot learning and multi-source training, where datasets from the target language are incorporated during training [17, 42, 55]. Model-level fusion is a concatenation of high-level hidden features within a model, with only specific layers or components sharing parameters. This is common in parallel model architectures like multi-channel learning [23], multi-task learning [29], and meta-learning [49, 50].

D. *Evaluating The Accuracy of HS Detection Models*

The accuracy of High-Score Detection (HS) detection models is evaluated using various metrics and methodologies. The confusion matrix, which includes True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN), is a key component. Accuracy is the percentage of correct predictions, calculated as $(TP+TN)/(TP+TN+FP+FN)$. Precision measures the accuracy of positive predictions, showing how many of the predicted HS instances are actually HS. Recall measures the model's ability to identify all relevant instances of HS. The F1-score, the harmonic mean of precision and recall, provides a balance between the two metrics, especially useful when dealing with imbalanced datasets.

IV. DETECTING MULTILINGUAL HATE SPEECH

In this section, In Table 1 we summarize studies that have focused on detecting multilingual hate speech where the training languages are the same as the target languages.

These studies review a range of techniques and models used to detect multilingual HS, with an emphasis on the importance of using deep learning, transfer learning, and data augmentation to improve performance in diverse linguistic environments.

TABLE I. KEY STUDIES IN MULTILINGUAL HATE SPEECH DETECTION – LANGUAGES AND METHODS USED

Ref	Year	Languages	Methods/ Results
[20]	2018	EN AND IT	(SVM) with linear and RBF kernels; Ensemble Models and Deep Learning methods .Results:The bakarov team achieved 84.4% accuracy in Italian and 70.4% accuracy in English using TF-IDF, SDV, and classification boosting models, outperforming the baseline AMI-BASELINE.
[29]	2019	En, Fr, Ar	The study utilized traditional and deep models like Logistic Regression, BiLSTM, and Sluice Networks to tackle linguistic issues like Arabic dialect variability and language mixing, with English excelling in sentiment and aggression type classification.
[24]	2019	En, Hi, and De	LSTM networks with word embedding input CNN base approach with Word2vec embedding Best system performance: English 0.78, Hindi 0.81, German 0.61. Most used approach: is LSTM networks with word embeddings.

[23]	2019	Es, GermEva, It	A study developed a multi-channel BERT model to detect hate speech across multiple languages using deep learning and transfer learning techniques. Data from Spanish, German, and Italian Twitter datasets was translated into English and Chinese using Google Translation API. The model outperformed most cases, with an accuracy of 82.2% on HaSpecDe and 79.8% on GermEval data.
[11]	2019	Hi, En, and Id	The study compares two machine learning methods for multilingual hate speech identification: translated and non-translated. The non-translated method offers better results but requires more data collection and annotation. Researchers combined the translated method with monolingual hate speech identification to improve performance.
[30]	2020	En, It, and Es	various machine translation models, including LSTM with monolingual word embedding, joint-learning models with multilingual word embeddings, HurtLex, BERT with multilingual model, and machine translation.
[52]	2020	En, Hi, and De	BERT variants, ALBERT, DistilBERT, RoBERTa, self-attention models F1 scores for task A: 0.51 (English), 0.53 (Hindi), 0.52 (German). F1 scores for task B: 0.26 (English), 0.33 (Hindi), 0.29 (German).
[26]	2022	En - Ar	A study developed a deep learning model for detecting offensive language in multilingual contexts using transformer models like BERT. Using a bilingual dataset, the model achieved an accuracy of 91.36% and an F1-score of 91%, outperforming other models like SVM and CNN. The translation-based method, particularly with AraBERT, outperformed the combined multilingual method.
[67]	2022	Ta ,En ,MI	The study used the Data Augmentation Technique (MTDOT) to generate offensive comments by translating them into intermediate languages and back into Tamil. The MuRIL model was used to convert texts into numerical vectors, and various classifiers were trained. The MTDOT outperformed SMOTE and Near Miss, achieving a 65% improvement in F1-Score. Multi-level translation generated more diverse data and higher quality, enhancing model performance.
[27]	2023	En, Es	The study evaluated deep learning models like LSTM, BiLSTM, CNN, and GRU individually and proposed an ensemble model to enhance performance. Word2Vec was used to represent words, and XGBoost was applied. BiLSTM performed best, while CNN and GRU were good but lagged. The ensemble model outperformed all models.
[15]	2023	EN, De, Hi, Ko Ar, Si, Es	The research paper introduces a pre-trained T5 model for detecting offensive language in multilingual texts, focusing on sentence-level and token-level classification tasks. The models are FT5, a pre-trained T5 model based on t5-large, and mFT5, a multilingual version of FT5 that supports languages like German, Hindi, and Korean. FT5 outperformed all other models in English, while mFT5 outperformed mBERT and XLM-R in six languages.
[33]	2023	It, De, Es	compared different models for detecting provocative speech across multiple languages, including German, Italian, and Spanish. Custom BERT models were used for each language, while cross-linguistic models such as mBERT and XLM-RoBERTa were used. XLM-RoBERTa was the best cross-linguistic model, achieving the highest MCC value of 0.56, outperforming mBERT. DistilBERT, a translation method, was used for translation into English when computing resources were limited. The study concluded that XLM-RoBERTa is the optimal model for detecting provocative speech across multiple languages.
[93]	2025	Hindi, Nepali (Devanagari script)	A hybrid model combining XLM-RoBERTa (for cross-linguistic embeddings) and BiLSTM with an attention mechanism The model achieved an F1-Score of 0.7481, outperforming:- XLM-RoBERTa alone (F1 = 0.6912).- Traditional models such as XGBoost (F1 = 0.6034) and Logistic Regression (F1 = 0.6528).

V. DETECTING HATE SPEECH ACROSS LANGUAGES

In this section, we summarize studies that have focused on detecting hate speech across languages and the transfer of knowledge from high-resource languages to low-resource languages, i.e., training languages are different from target languages.

Zero-shot HS detection is challenging due to limited labeled data in languages. Transformer-based models excel in this context, especially when trained on diverse datasets. Natural language inference models show performance improvements across multiple languages. Transfer language selection is crucial for effective zero shot cross-lingual abusive language detection, especially for low-resource languages. Utilizing existing datasets from higher-resource languages and advanced architectures like frozen Transformer models can optimize performance in zero-shot settings. Nozza's cross-lingual transfer learning framework

for detecting HS on social media shows lower performance in Spanish and Italian compared to English, highlighting the need for context-aware models [44]. Lukas Stappe et al. developed a method for detecting HS using frozen transformers and AXEL, achieving competitive results in English and Spanish subgroups. The AXEL attention-based classification block significantly improved HS detection performance. However, the study's limitations include focusing on English and Spanish, not exploring low-resource languages, and not discussing scalability and efficiency [43]. Montariol et al. propose a multilingual architecture for training auxiliary tasks in both source and target languages, enhancing HS detection capabilities. They found some HS labels more susceptible to cross-lingual transfer learning [41]. The study by Jet et al. evaluates the effectiveness of transfer language selection for zero-shot cross-lingual abusive language detection using datasets from seven languages across three language families. The results show that the multilingual transformer model XLM-R outperforms mBERT in abusive language detection tasks, with Germanic and Slavic languages achieving the highest scores [40]. The study by yzhova A et al. focuses on fine-tuning a model for detecting HS in various languages, using toxic comments in Russian and Hindi tweets as training datasets. The model is tested on English and Russian religious texts, and a text attack algorithm is applied. Cross-lingual Transformer models are explored for processing short text messages and identifying intolerance. The XLM-R model achieves high HS detection accuracy, but basic adversarial attacks decrease its accuracy [39]. Muti et al. employ monolingual and multilingual transformer models to detect misogyny in tweets across English, Italian, and Spanish using a transfer learning approach. Single-language models yield the best results, but zero-shot classification decreases. Error analysis reveals common misogyny detection pitfalls.[38]. Laza-del Arco FM et al. study uses zero-shot learning (ZSL) with prompting to detect hate speech (HS) across multiple languages, addressing issues like limited labeled data and HS variability. They use large language models and prompt formulations to classify text as hateful or non-hateful, highlighting the importance of selecting appropriate verbalizers.[12].Bigoulaeva et al. train neural network models in English and German using cross-lingual word embeddings, addressing data scarcity and label inconsistencies. They use CNN, BiLSTM, and multilingual BERT architectures, fine-tuned on source and target languages, and data sampling techniques to address class imbalance.[31].Juru SS et al. examines HS detection in 9 languages using 16 sources, focusing on English datasets. It finds simple models like LASER embedding and logistic regression outperform complex ones in low-resource settings while BERT-based models are more effective in high-resource meta-learning settings., and zero-shot classification is effective in Italian and Portuguese languages [13]. Rodríguez SE et al. uses Language Agnostic BERT Sentence Embeddings (LabSE) to detect HS in mono-lingual, multi-lingual, and cross-lingual settings. It tested models like Support Vector Machines and Tree-based models, and representations from Multi-lingual BERT. LabSE improved performance in mono-lingual and cross-lingual settings, and its multilingual contextualized representations can be used across models and transfer levels [51] Mozafari et al. study explores meta-learning in cross-lingual HS detection, focusing on low-labeled data. They use Model-Agnostic Meta-Learning (MAML) and Proto-MAML models. Results show meta-learning low-labeled models outperform traditional transfer learning models, with Proto-MAML being the best performer. [50]. Zhou L et al. study cross-cultural transfer learning for detecting offensive language in Chinese, focusing on Korean and English cultures. They found that cultural biases affect model adaptability, and using intra-cultural and cross-cultural training sets, multilingual LMs were applied, fine-tuned for 100 epochs, and explored cultural differences in translated data settings.[57]. Firmino et al. proposes a novel method for detecting HS on social media using Cross-Lingual Learning. The method uses transfer learning from Pre-Trained Language Models (PTLM) to solve problems in languages with fewer resources. The methodology involves four stages:

corpora acquisition, PTLM definition, training strategies, and evaluation. Experiments were conducted using English, Italian, and Portuguese Pre-Trained Language Models, with the OffComBr-2 corpus being the target language. The results showed promising results, with the OffComBr-2 corpus showing the best state-of-the-art result [46]. Koyel Ghosh et al. used transformer-based language models on Indian languages, creating Assamese and Bodo hate speech datasets. They conducted cross-linguistic experiments on Hindi-Marathi, Hindi-Bodo, Marathi-Bodo, and Bengali-Assamese. The results show that monolingual training ensures superior performance only if the raw data is sufficient, while the initial training is performed. The multilingual models performed best on Hindi, Bengali, and Bodo datasets, while the monolingual models outperformed on Marathi and Assamese[68]. Juuso Eronen et al. have developed a transfer language selection approach to detect abusive language across languages using linguistic similarity measures to identify ideal source languages, and improving classifier performance for low-resource languages by leveraging data from higher-resource languages, rather than relying solely on language family intuition [69].

TABLE II. KEY STUDIES IN DETECTING HATE SPEECH AND TRANSFERRING KNOWLEDGE ACROSS LANGUAGES – LANGUAGES AND METHODS USED.

Ref	Year	Languages	Methods and Transfer Strategies
[25]	2019	En, Fr	BiLSTM using multilingual embeddings, including Glove and FastText word vectors Results: The best feature model is BiLSTM + Attention, which outperforms neural models due to small data size. The multilingual detection is performed using FastText + BiLSTM, with a 79% accuracy. Common errors include missing context, using stereotypes or tropes, irrelevant data, ethical bias, and a lack of data, particularly for sexism in French.
[53]	2020	Ar, Da, En, El, Tr	Multilingual BERT, and augment training data with translation. Transfer learning from English to Danish using the few-shot strategy. Results: The study found that model accuracy increased to 85% F1 for Greek and 84% F1 for Danish through transferable sampling, and performance improved with knowledge transfer from English to French, despite limited data. However, the models rely on machine translation, struggle to distinguish non-hostile contexts, and mostly focus on European languages.
[22]	2021	En, Es, It Evalita2018 IberEval2018	model (CCNL-Ex) uses a bilateral architecture for text processing, linguistic embeddings, feature extraction, capsule network, and Softmax for classification. It uses HurtLex and Negative Sentiment Lexicon for offensive words. (Translate the target language into the same source language) Results: The study suggests that combining capsule networks with specialized dictionaries can improve hate speech detection performance across languages, but also calls for improved machine translation and expanded model inclusion.
[63]	2021	Eng, Fr, De, Id, It, Pr, Es	Machine Translation (MT) using Google Translate API. Joint-learning models using multilingual language representations Incorporating information from a multilingual lexicon of abusive words. Results: Joint-learning models achieve best performance on most languages. Machine translation and pre-trained English model achieve robust performance.
[55]	2022	En, Fr, De, Es	Multilingual Transformer Models (XLM-RoBERTa); translation (Gathering human-generated data for code-switched text. Creating synthetic code-switched offensive content data.) Results: Human-annotated test sets were generated for the resource-poor en-fr, en-de, and en-es language combinations, and we proposed an algorithm for generating synthetic code-switched data for training in resource-poor domain
[54]	2022	En, Ne, Si, Bn, Ta, Sw	Unsupervised Learning; Transferring learning; Cross-lingual offensive speech identification Results: unsupervised attack detection method for identifying cross-lingual offensive languages in networks with numerous offensive languages, transferring rich and "false" low resource language data sets to a model.
[45]	2020	En-Es	Transformers Models (BERT, XLM) - Translation (zero-shot - FEW-shot) frozen, pre-trained Transformers for hate speech detection, along with an AXEL attention-based classification block for enhanced results. Results: AXEL (F) is best at handling multiple languages even without dedicated training (zero-shot). The differences in performance across languages confirm that the underlying architecture varies by language. Translation improves performance when used in training and testing.
[42]	2022	Bn, En, Hi, Ur, (Ta-En), (Ma-En), (Ka-n), Mr	Zero Shot - Translation - Zero Shot Little using m-BERT, MuRIL Results: They found MuRIL outperforms m-BERT in 7 out of 10 language types, and zero-shot model transfer is effective for abusive language detection. AIBOne is most effective for fewshot settings, and synthetic silver instances are helpful for low-resource languages. However, performance decreases against adversarial attacks

Ref	Year	Languages	Methods and Transfer Strategies
[69]	2022	Eng, De, Da, Pl, Ru, Ja, Ko	Cross-lingual transfer learning and Linguistic similarity metrics Results: Linguistic similarity significantly influences model performance, with Russian excelling as transfer language, even for unrelated languages like Japanese and Korean, while English generally performs poorly.
[70]	2022	Eng, Es, It, De, Ar, El, Tr	The process involves training a base model using XLM-RoBERTa, a multilingual model labeled with English data, and generating pseudodata for the target language, such as Spanish, using the base model. The final model, a monolingual model like Spanish BERT, is then trained to improve performance. Results: The proposed model significantly improved in all languages, with .46% improvement in Greek, and significantly improved in classifying hidden expressions like "puta" (Spanish) and "porca" (Italian).
[8]	2022	En and De	The study aims to enhance the detection of hate speech in low-resource languages using cross-linguistic transfer learning techniques. Using German as a target language and English as a source language, models like CNN, BiLSTM, and mBERT were proposed. The mBERT model performed best, with precision of 52.29% for "Hate" and recall of 93.30% for "noHate". Transfer learning is effective without pre-labeled data.
[37]	2023	It, metalearningEs	Zero-shot learning with prompting for hate speech detection (BERT, XLM-R, FLAN-T5, mT0, RoBERT); Results: Zero-shot with prompting shows promising results for hate speech detection.
[49]	2023	En, Es, De, Hi, It, Ar, Da, El, Tr	create a meta-learning model that can quickly adapt to new languages using limited data, using Model-Agnostic Meta-Learning (MAML, MAML based mBERT and XLM-R (zero-shot)). Key components include meta-training, self-refinement, and domain adaptation loss, which improve the model's generalization across different domains. Results: The model has strengths such as resource efficiency, flexibility, and domain bias handling. It can adapt to new languages with minimal data and works with any language model. However, it has weaknesses such as computational complexity, reliance on silver labels, and poor performance when transferring knowledge between different language families.
[36]	2023	En, De, zh	Transformer models and multilingual embedding (LASER+SVM, LASER+LR, BERT+translation Mbert) Results: Transformer models outperformed SVM models in all experiments, especially for monolingual models. German BERT achieved a 0.14 MCC difference, while multilingual models like XLM-RoBERTa had the highest detection performance. DistilBERT, with translated data, achieved results close to XLM-RoBERTa, making it suitable for limited computing resources. Multilingual models were best for German, Italian, and Spanish, while monolingual models provided good results.
[28]	2024	Eng, De, Hi	Semi-supervised GAN model combining mBERT and LM-RoBERTa; Generative Adversarial Networks (GANs) and Pretrained Language Models (PLMs)
[14]	2024	En, Es, It	Use Hurltlex Dictionary; Monolingual and multilingual translation word embedding (mBERT, MUSE, HateEmb, LASER, XLM) Results: The study reveals that hate speech is expressed in common patterns across languages, such as neutral words in hostile contexts. It outperforms in four out of six settings, including English → ES and EN → IT. However, it has limitations, including reliance on binary dictionaries and testing on three European languages.

We conclude from previous studies Recent studies in cross-lingual hate speech detection demonstrate that Transformer-based models like BERT and XLM-R achieve promising results in low-resource settings, particularly when using zero-shot or few-shot learning. However, the performance of these models varies depending on source language selection, linguistic similarity, and the availability of diverse datasets. Challenges remain, such as decreased accuracy against adversarial attacks, cultural biases, and difficulties in knowledge transfer to low-resource languages. To enhance performance, researchers propose advanced techniques like pseudo-label fine-tuning, multi-task models, and auxiliary pivot languages. Ultimately, these studies highlight the need for more context-aware models that account for cultural and linguistic variations to improve hate speech detection across different languages.

VI. DATASETS ON MULTILINGUAL HS DETECTION

Through the survey of previous studies to detect multilingual HS, we found that the first step to structuring the detection of HS is to collect data and create data, as there is no ready data set and a set in one file from several languages, so researchers in this field created their own data set by searching for a set or datasets for each language separately and then creating a data set. There is no standard data set to refer to, and this is considered a gap because the data set that is created multilingually must have linguistic diversity in HS and cultural and balanced in the number of

specific examples. The number of examples that do not incite HS. Through the survey we conducted, we noticed that the number of examples that indicate HS is much less than the examples that do not incite hate. One of the challenges we also face is the categories in each data set, as each data set is divided into categories that are different from each other. For example, there is a data set consisting of two categories (hate and not hate) and a data set consisting of three categories such as (normal - offensive - hate) and so on. When researchers created their own data set, they used strategies to unify the categories. The majority followed the method of forming categories of two types (hate - normal) and ignored the other categories, and others made the category of categories that indicate abuse, alienation and hatred as if they were all HS. Multilingual lexicons, like HurtLex, are word-aligned dictionaries that provide translations or equivalents of terms across multiple languages[4]. One of the studies that emphasized the necessity of linguistic diversity and domain diversity of offensive language and its role in cross-linguistic transmission to detect HS is a study in which the authors presented the XHate-999 multi-domain and multilingual evaluation dataset to detect offensive language. They concluded through their results that the results are good if the source and target languages are from the same domain, and on the contrary, they achieve results that are not good if the domains of the source and target languages differ [34]. Muhammad, et al presented the first identification collection of hate speech and abusive language in African languages, Twitter data called Afrihath[92]. Table 3 shows the most important datasets. Multilingual HS datasets face challenges due to data scarcity, linguistic diversity, and complexity. Existing models underrepresent low-resource languages, requiring meta-learning techniques for adaptation. The subjective nature of HS complicates data collection, requiring careful curation and labeling [49].

TABLE III. TOP DATABASES FOR DETECTING MULTILINGUAL HATE SPEECH

Reference	Year	Languages	Size	Source
[29]	2019	Ar, En, Fr	En: 5647, Fr: 4014, Ar:3353 (Total 13,014)	Twitter
[72]	2023	Malay	4892	Twitter
[73]	2019	Tunisian (AR)	6,075 (HS and abusive platforms)	Different social media
[71]	2019	En	19,600	Twitter
[74]	2019	Id	5,561 (abuse, HS)	Twitter
[75]	2019	It	5,668 (HS, offense)	Twitter
[66]	2020	Tr	36,232	Twitter
[76]	2020	Da	3,600 (abuse, offense)	Facebook,Reddit, Twitter
[77]	2020	It	12,081	Immigration, Roma, Muslims
[78]	2020	En	14,949(HS)	Twitter
[79]	2019	En, Es	En: 12,805, Spanish: 6600 HS	Twitter
[80]	2019	Pt	5,668	Twitter
[81]	2019	Ar (Levantine)	5,846	Twitter
[82]	2019	Es	6k (6000) HS	Twitter
[65]	2022	Chinese(COLD dataset)	37,480	social media Zhihu and Weibo
[83]	2022	Korean	40,429	NAVER news and YouTube
[84]	2019	En, Fr, and It	4078	Twitter
[85]	2020	En	9,000,000	Twitter
[86]	2019	Bengali (BN)	5,126	Facebook
[87]	2024	Roman Urdu	20,000	Twitter
[88]	2021	En	41,255	Synthetic
[89]	2024	AR (Jordanian)	557,551	Twitter
[90]	2020	Roman (Ur)	10,012	Twitter
[91]	2020	Ko	9,381	A popular domestic entertainment news aggregation platform

[32]	2018	En	27,330	Twitter
[92]	2025	Afrihate dataset Ary,Arg,Hau,Ibo,Kin,Oro, Pen,Som,Swa,Tir,Twi,Xho, Yor,Zul	172,780 (Hate + Abusive + Neutral) Ary(1,250),Arg(4,634),Hau(6,644),Ib o(5,003),Kin(4,722),Oro(5,032),Pen(1 0,599),Som(4,660),Swa(21,092),Tir(5 ,072),Twi(3,901),Xho(3,683), Yor(4,879),Zul(4,308)	Twitter

VII. CHALLENGES AND LIMITATIONS

In this section, the challenges faced by researchers in detecting multilingual hate speech are reviewed, which are:

The challenges of multilingual HS datasets are multifaceted, primarily revolving around data scarcity, linguistic diversity, and the complexity of HS itself. These issues hinder the development of effective detection mechanisms across various languages: The scarcity of annotated data across languages is a significant challenge in building accurate and per-formant models. This data is often non-accessible and requires extensive expertise. Traditional machine learning techniques often perform poorly on minority class samples, making it difficult to create accurate models. Researchers have suggested oversampling and under sampling strategies to address this issue. Even non-English datasets present challenges, as there are limited annotated datasets in the domain. Therefore, creating annotated datasets in multiple languages is crucial for advancing approaches, especially in low-resource languages. The correlation between the languages used as source and target languages in the multilingual detection of hate speech.

Cross-language learning is challenging due to limitations in detecting HS across languages, particularly with culturally specific swear words. Current machine translation tools struggle to translate these words, and some methods perform inconsistently across different target languages, highlighting linguistic issues. Biases, including racial, author, and subject biases, can hinder the recognition of HS in multiple languages, with subject bias being the most important issue.

HS tasks can be influenced by cultural backgrounds, dialects, and the definition of HS. Different jurisdictions provide different definitions, making them more complex in multilingual scenarios. Multilingual text classification requires extensive computational resources due to large data volumes and complex models. Pre-trained language models, such as multilingual LLMs, have limitations, such as the” curse of multilingualism” and the” curse of multilingualism.” Machine translation tools, such as multilingual Neural Machine Translation (NMT), have limitations in handling many languages and often give low performance compared to bilingual models.

Multilingual pre-trained language models (PLMs) face challenges in generalization abilities, overfitting, cross lingual transfer performance, and interpretability. Transformer-based multilingual PLMs can be inconsistent, leading to bias and instability. Machine translation can help augment datasets but can also experience performance degradation. Interpretability is poor, making it difficult for humans to understand model performance.

VIII. RESEARCH DIRECTIONS

The survey reveals that most research has focused on cross-linguistic transfer using zero-shot and few-shot strategies, pre-trained language models (LLMs), and specific language families. The authors propose developing datasets for detecting hate speech in low-resource languages like Arabic, Persian, Nigerian, Hebrew, Kurdish,and African languages[92]. There is a need for comprehensive datasets, improved data annotation strategies, and text generation techniques to increase data volume. Pre-trained language models should be used without additional training to optimize computational costs. Strategies for selecting the appropriate transfer language and employing linguistic similarity techniques are being explored. Language training strategies, including adaptive techniques, could bridge the gap between hate speech detection and programming language models. Despite progress made by LLMs,

challenges remain, such as biases in data and difficulty in generalizing models across different languages and cultures. The study primarily examined traditional transformer models and large language models like BERT, but the rapid adoption of large autoregressive models like ChatGPT, Gemini, and Claude for multilingual hate speech detection necessitates further research to evaluate their performance, robustness, and applicability.

IX. CONCLUSION

The survey emphasizes the need for cross-linguistic models to detect hate speech in unseen or low-resource languages. Despite advancements in large language models (LLMs), challenges such as data biases and difficulty generalizing models across languages and cultures persist. Effective strategies include transfer learning, multi-task learning, and meta-learning for transferring knowledge from high-resource languages to low-resource languages. Transformer based models like BERT and XLM-RoBERTa show superior performance in detecting hate speech across multiple languages. However, strategies to determine the language of transfer are still needed. Significant research gaps remain, such as the need for balanced datasets, improved model interpretability, and reduced data biases. The research highlights promising opportunities for future research in multilingual hate speech detection, emphasizing the need for robust models and collaboration among researchers.

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